

ESS Simulation based on PV Generation Prediction

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Abstract—For the renewable energy business, technologies that can maximize energy efficiency such as energy information collection, energy management and operation, and economic analysis are needed. You need to be able to calculate thorough economic analysis and profit before you need it. It is necessary to be able to analyze the present energy demand to predict the future demand, and to be able to calculate the energy production requirement based thereon. In this way, it is necessary to be able to estimate the facilities considering economic efficiency and calculate the economic benefit for each region. For this, this paper shows ESS Simulation based on PV Generation Prediction.

Keywords— *ESS, energy management, energy big data, energy information collection*

I. INTRODUCTION

Renewable energy, including photovoltaic energy, has the disadvantage that it can not be produced in a timely manner when needed compared to energy using fossil fuels. In the case of solar energy, it reacts sensitively to seasonal weather changes, and it can be produced in the daytime, but it is time constrained as production is impossible at night. Despite the limitations of such production, interest in renewable energy development is increasing due to environmental problems in recent years. At the government level, it is a tendency to allocate a lot of intensive to the renewable energy generation companies and to support the expenses for the facilities.

In order to operate energy efficiently, a lot of data related to energy production, storage and use should be collected and managed. Popular Big Data technology is a good solution for collecting and storing various kinds of data. However, for energy management, it is necessary to establish an organic relationship between these data. You need to be able to define and associate data relationships by time, space, and context.

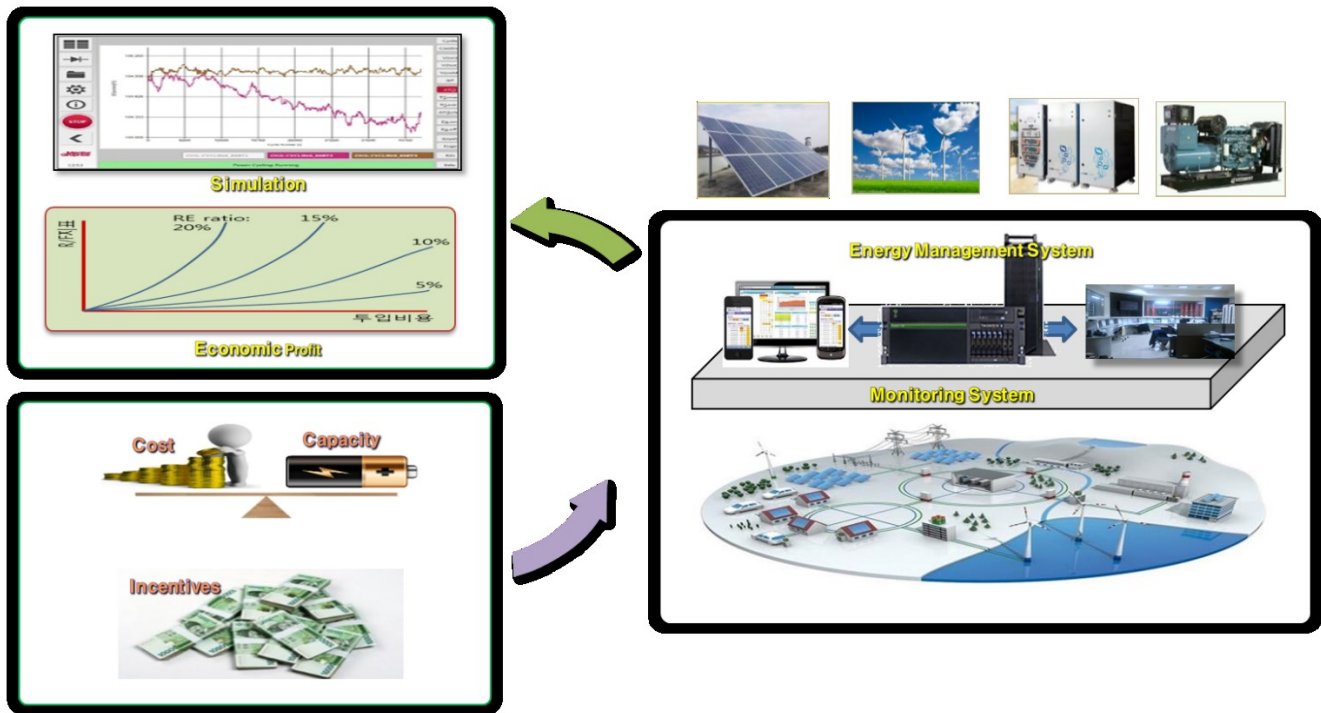
As interest in renewable energy increases, individuals are increasingly interested in renewable energy projects. However, the new and renewable energy business is not yet profitable compared to the cost of equipment. This requires an efficient energy management based on a deep

understanding of renewable energy. It is necessary to be able to analyze the present energy demand to predict the future demand, and to be able to calculate the energy production requirement based thereon. In this way, it is necessary to be able to estimate the facilities considering economic efficiency and calculate the economic benefit for each region. For this purpose, this paper introduces the energy capacity simulation technique through prediction of renewable energy and solar energy generation.

II. PV GENERATION PREDICTION

There are three main types of information that affect energy. Data affecting energy production, data affecting energy consumption, and data on energy storage and management. The data that affect energy production are climate information, production site information, and production time. Data that affect energy consumption are consumer information, consumption patterns, consumer device information, and climate. Data on energy storage and management are storage device information such as ESS, operating policy, power unit price, and power contract information. It is the purpose of the new and renewable energy to send the operation command such as command / stop, use, storage, etc. of new and renewable energy by analyzing this information well.

PV generation is influenced by various factors. Weather data such as temperature and cloudiness are the most important items, but the influence of the altitude of the location where PV is installed and surrounding buildings is also important. In order to predict PV generation, periodic collection and forecast data of weather data should be reflected immediately. In addition, past generation data can provide good information for prediction. If there is insufficient data in the same area, historical data of similar conditions is also important. To do this, we need to periodically collect data from various aspects and elements based on the Big Data System. In this paper, we have constructed a big data platform for accurate PV generation. The measured data collected per second can be preprocessed and informed and used in the prediction engine to estimate the PV generation amount after 24 hours.



[Fig 1] Energy Management System based ESS Simulation

III. ESS SIMULATION MECHANISM

In order for renewable generation to be serviced, it must be possible to calculate how much ESS and PV should be installed in the service area. This is based on PV generation forecasts of how much solar energy can be produced in this region. Based on the predicted PV power generation, it should be decided how much ESS should be installed in the service area. Of course, it would be nice to install a lot of Ess with a lot of capacity, but now the cost of ESS is very high. Because it is a consumable item, it is necessary to be able to calculate the most reasonable ESS capacity considering economical efficiency because the replacement cost is large when it is used over a certain cycle. For this, price information of ESS that is sold on the market is needed.

It should be able to calculate the rational ESS capacity based on the generation pattern of the service area by analyzing the capacity information, device life information and characteristic information. The price information and the device characteristic information are volatile information, and the information should be periodically updated. To this end, the system periodically collects and manages service area characteristic information, ESS price information, and device characteristic information. Based on this information, we calculated the ESS capacity which is most suitable for PV power generation.

Based on the calculated ESS capacity, it is simulated based on predicted hourly power generation information.

The ESS SOC was simulated by considering the characteristics and life span of the ESS given the climatic conditions to determine whether one ESS is good or not and how many ESSs should be used. Economic benefits. We present the largest case of economic benefits and present the most appropriate ESS capacity and quantity for the service area.

ESS system construction ESS capacity and operation policy are determined according to the optimal scenario considering the economics from the viewpoint of the operator. Based on the algorithm for implementing the basic functions of the simulator, the concept of compensating the error between the KPX bid amount and the actual generation amount is applied to the ESS by using the past generation data and the predicted generation amount data to calculate the ESS capacity. - Minimize ESS fluctuations by bidding at a median of the evolution estimates that change over an hour. If there is no previous generation data, data is generated by adding random error to the generation forecast. Based on the bid amount, we calculate reliability / flexibility based on the error data that occurred in ESS operation. The PV operational information collection and control interface defines the data models needed for grid operation and applies them to data from distributed sources.

IV. CONCLUSION

In this paper, we describe how to calculate the ESS capacity through the prediction of PV power generation, which is a necessary technology for estimating the facilities needed for the renewable energy business. In order to calculate the ESS capacity, it is necessary to be able to present the most economical capacity considering the characteristics and life span of the ESS, reflecting the operational characteristics of the ESS rather than simply numerical calculation. To do this, we simulate the ESS by time zone using the predicted PV generation information and apply the various ESS by capacity to propose the most economical ESS capacity. If we simulate based on more diverse factors affecting future renewable energy generation, it will be possible to calculate more accurate ESS capacity.

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